



Mammographic Mass

Donated on 10/28/2007

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Creators

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DOI

10.24432/C53K6Z

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Discrimination of benign and malignant mammographic masses based on BI-RADS attributes and the patient's age.

Dataset Characteristics

Multivariate

Subject Area

Health and Medicine

Associated Tasks

Classification

Feature Type

Integer

Instances

961

Features

5

Dataset Information

Additional Information

Mammography is the most effective method for breast cancer screening available today. However, the low positive predictive value of breast biopsy resulting from mammogram interpretation leads to approximately...

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Has Missing Values?

Yes

Introductory Paper



[The prediction of breast cancer biopsy outcomes using two CAD approaches that both emphasize an intelligible decision process.](#)

By M. Elter, R. Schulz-Wendtland, T. Wittenberg. 2007
Published in Medical Physics (Lancaster)

This allows for the sharing and adaptation of the datasets for any purpose, provided that the appropriate credit is given.

Variables Table



Variable Name	Role	Type	Demographic	Description	Units	Missing Values
BI-RADS	Feature	Integer				yes
Age	Feature	Integer	Age			yes
Shape	Feature	Integer				yes
Margin	Feature	Integer				yes
Density	Feature	Integer				yes
Severity	Target	Binary				no

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Additional Variable Information

6 Attributes in total (1 goal field, 1 non-predictive, 4 predictive attributes)

1. BI-RADS assessment: 1 to 5 (ordinal, non-predictive!)
2. Age: patient's age in years (integer)
3. Shape: mass shape: round=1 oval=2 lobular=3 irregular=4 (nominal)
4. Margin: mass margin: circumscribed=1 microlobulated=2 obscured=3 ill-defined=4 spiculated=5 (nominal)
5. Density: mass density high=1 iso=2 low=3 fat-containing=4 (ordinal)
6. Severity: benign=0 or malignant=1 (binominal, goal field!)

Missing Attribute Values:

- BI-RADS assessment: 2
- Age: 5
- Shape: 31
- Margin: 48
- Density: 76
- Severity: 0

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Dataset Files

File	Size
mammographic_masses.data	13.1 KB

File	Size
mammographic_masses.names	3.3 KB

Papers Citing this Dataset



SORT BY YEAR, DESC

[Variational Langevin Hamiltonian Monte Carlo for Distant Multi-modal Sampling](#)

By Minghao Gu, Shiliang Sun. 2019

Published in ArXiv.

[Falling Rule Lists](#)

By Fulton Wang, Cynthia Rudin. 2014

Published in ArXiv.

[Reliable Probabilistic Prediction for Medical Decision Support](#)

By Harris Papadopoulos. 2011

Published in EANN/AIAI.

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Reviews



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